

A photograph of a charcoal grill sitting on a wooden table outdoors. The grill is filled with glowing charcoal briquettes. The background shows a grassy area and some trees. The entire image is framed by a green border.

TECHNICAL OCCUPATIONAL MECHANICAL TECHNOLOGY WELDING

Learner's Guide

Year 3 / Term 1

**[https://www.youtube.com/watch?
v=F96IUTMkYHY](https://www.youtube.com/watch?v=F96IUTMkYHY)**

INTRODUCTION TO MECHANICAL TECHNOLOGY: WELDING

2.1 What is Welding?

Welding is a fabrication process that joins materials, usually metals and other materials. This is often done by melting the work pieces and adding a filler material to form a pool of molten material that cools to become a strong join.

Metalworking is the process of working with metals to create individual parts, assemblies, or large scale structures.

2.2 Topics to be studied in Welding

1. Safety and first aid – Occupational Health and Safety (OHS) Act;
2. Workshop environment
3. Drawings (rough sketches, simple drawing ISO, Ortho and CAD welding related)
4. Tools and equipment (hand, power and machine)
5. Materials – ferrous and non-ferrous materials
6. Joining methods – permanent and semi-permanent.
7. Welding (operations/procedures and equipment).

2.3 **Specific Aims:**

The learner is able to:

1. Adhere to and identify safe working practices and demonstrate safe working conditions daily, also adopting proper safety and first aid procedures.
2. Demonstrate knowledge of the welding industry and its productivity requirements, by applying appropriate work-procedures.
3. Understand and interpreting work instructions and drawings for the completion of projects.
4. Apply welding technology, techniques, processes and skills, as applied in the fabrication and welding industry, using appropriate tools and measuring equipment.
5. Identify Ferrous & Non-Ferrous metals the differences, applications and uses of them.
6. Demonstrate the different welding processes: Arc (AC/DC), MIG, TIG, Brazing, Soldering and spot welding.

Demonstrate understanding of the following:-

What Occupational Health and Safety requirements do companies have to comply with in South Africa?

In South Africa, companies are required to comply with the [Occupational Health and Safety Act, 85 \(OHSA\) of 1993](#). This act establishes the rights and duties of both employers and employees with regard to health and safety in the workplace.

Under the OHS Act, employers are required to provide a safe and healthy working environment for their employees. This includes taking measures to prevent accidents and injuries, as well as providing necessary protective equipment and ensuring that all machinery and equipment is in good working order. Employers are also responsible for providing training and information to their employees about potential hazards and how to avoid them.



SAFETY PROCEDURES

Safe Operating / Safe work Procedures should be put together for workplaces. This aids workplaces to function in an organized fashion, these are documented safety guidelines for procedures and tasks to ensure that activities in the workplace are conducted safely. All activities must abide to these guidelines. Without these protocols, incidents and injuries would be the order of the day.

BEFORE SAFE OPERATING PROCEDURES CAN BE IMPLEMENTED, THE FOLLOWING STEPS HAVE TO BE TAKEN:

Identify the [hazards](#) and the risk associated with them - Risk Assessment

Evaluate the hazards involved when performing a task

Establish what personal protective equipment is required,

Document the steps required so that an activity is completed without any incidents.

The importance of Safe Operating Procedures is that they provide the tools for

7. Apply a variety of welding, oxy-fuel cutting and oxy-fuel joining processes.